Radial Basis Function (RBF) Kernel

$$e^{-\frac{1}{2}(a-b)^2} = e^{-\frac{1}{2}(a^2+b^2-2ab)} = e^{-\frac{a^2}{2}} e^{-\frac{b^2}{2}} e^{ab}$$

$$e^{ab} = 1 + \frac{ab}{1!} + \frac{(ab)^2}{2!} + \frac{(ab)^3}{3!} + \frac{(ab)^4}{4!} + \frac{(ab)^5}{5!} + \ldots\ldots\infty$$

$$e^{ab} = \begin{bmatrix} 1 & \frac{a}{\sqrt{1!}} & \frac{a^2}{\sqrt{2!}} & \frac{a^3}{\sqrt{3!}} & \frac{a^4}{\sqrt{4!}} & \ldots & \ldots & \infty \end{bmatrix}$$

$\phi(a)^T$

$\phi(b)$

$$\begin{bmatrix} 1/b \\ \sqrt{1!}/b^2 \\ \sqrt{2!}/b^3 \\ \sqrt{3!}/b^4 \\ \sqrt{4!}/\ldots \\ \ldots \\ \ldots \quad \infty \end{bmatrix}$$

Taylor series expansion of (x)

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \ldots\ldots\infty$$

$$e^3 = 1 + \frac{3}{1!} + \frac{3^2}{2!} + \frac{3^3}{3!} + \frac{3^4}{4!} + \frac{3^5}{5!} + \ldots\ldots\infty$$

Approx. to 4 terms	⇒	**13**
Approx. to 5 terms	⇒	**16.375**
Approx. to 6 terms	⇒	**18.4**
Approx. to 7 terms	⇒	**19.413**
Approx. to 8 terms	⇒	**19.84**
Approx. to 9 terms	⇒	**19.8420.01**
Actual Value	⇒	**20.086**